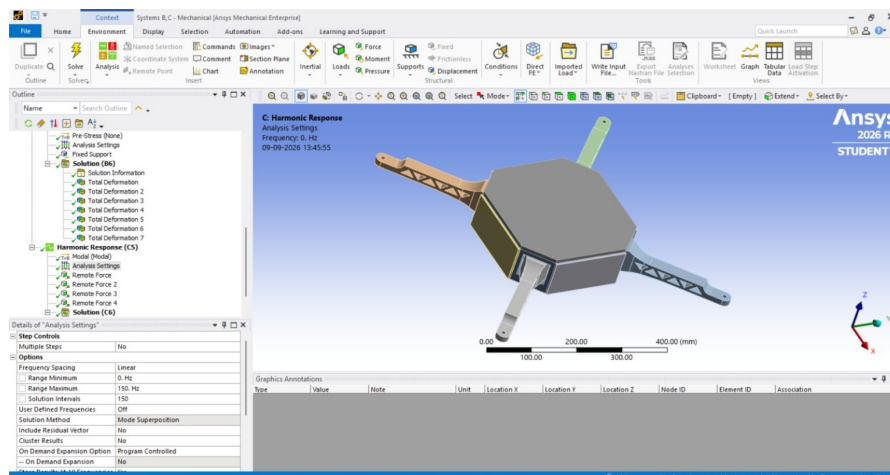


8. Modal Analysis

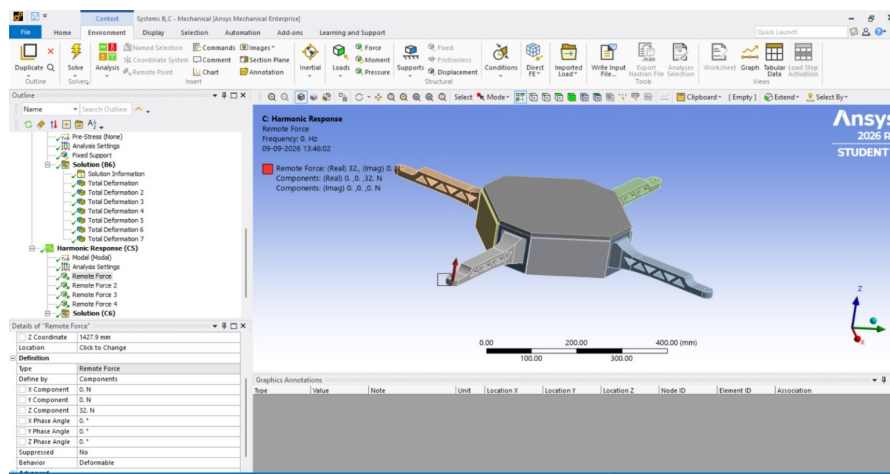
The modal model retains the structural geometry and fixed support while replacing the detailed motor/propeller geometry with four point masses of 0.105 kg at the motor locations. Six natural frequencies were extracted.

Mode	Natural frequency
1	72.582 Hz
2	72.605 Hz
3	72.618 Hz
4	72.620 Hz
5	89.963 Hz
6	89.971 Hz

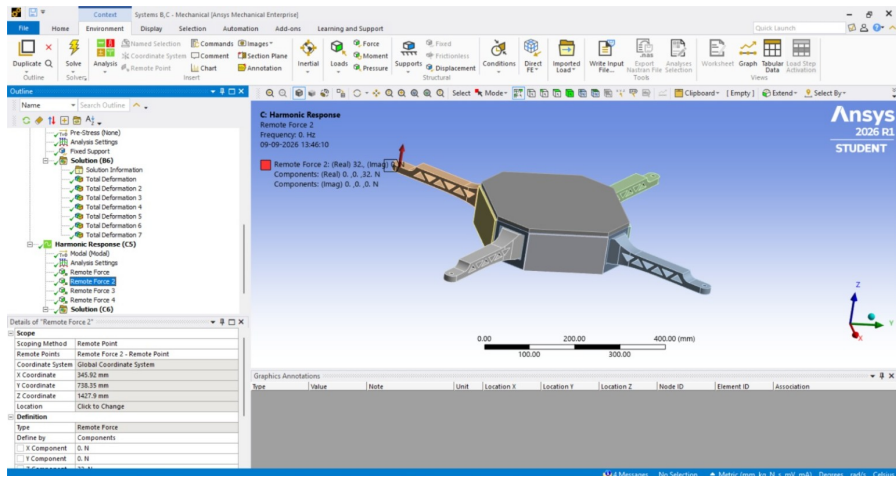
The first four modes are tightly grouped around 72.6 Hz, reflecting the near-symmetric four-arm architecture. A second pair occurs at approximately 90 Hz. The outer arms and motor regions dominate the relative mode shapes.



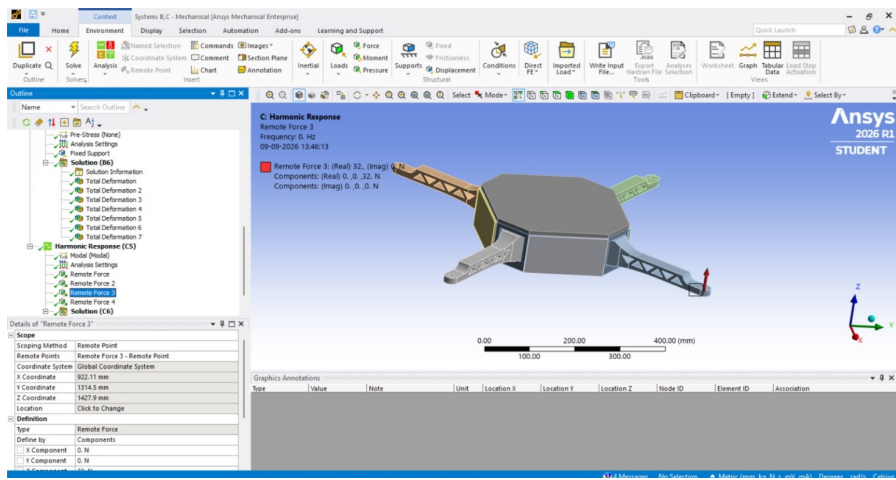
Modal deformation — Mode 1 region.



Modal deformation — Mode 2 region.



Modal deformation — Mode 3 region.



Modal deformation — Mode 4 region.